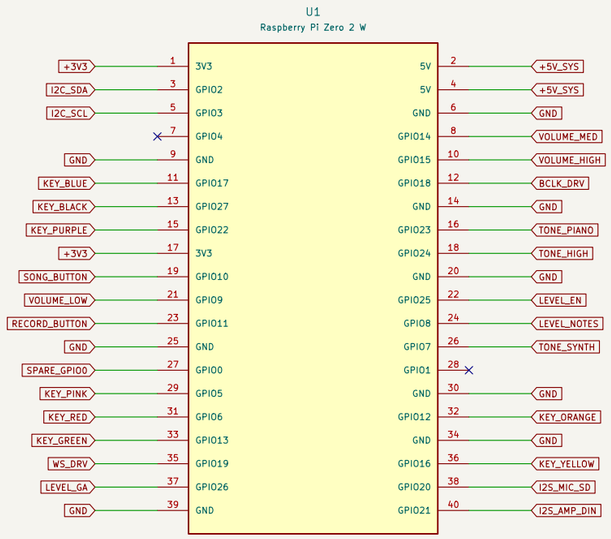
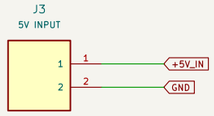
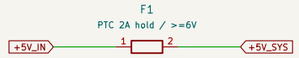
PI HEADER — BCM assignments preserved



U1: Pi Zero mechanical outline + mounting holes.
5 V enters GPIO header; 3.3 V is supplied BY the Pi.
No HAT EEPROM. GPIO1 / pin 28 remains unused.

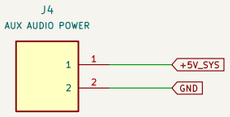
REGULATED 5 V INPUT

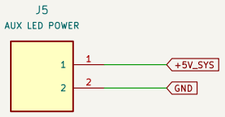


















Keyboard, buttons and Random LED

File: Controls.kicad_sch

Tone buttons, level and volume

File: Selectors.kicad_sch

I2S audio modules and AS5600 connector

File: Audio_Sensor.kicad_sch

USB supply +5 V -> OFF-BOARD SPST SWITCH -> J3 pin 1.
USB supply GND -> J3 pin 2 (ground is not switched).
Pi USB power stays disconnected. Turn switch off after OS halt.
No reverse-polarity or supply-ORing circuit.
Pi, amp and auxiliary loads share F1/current budget.
Verify supply, switch, connector and wire ratings.

Rev A0.2: pixel omitted; microphone gets BCM20.
Do not apply the PCB pinmux/amp changes to the existing toy yet.

SOURCE-BASED FOOTPRINTS — PHYSICAL FIT CHECK REQUIRED

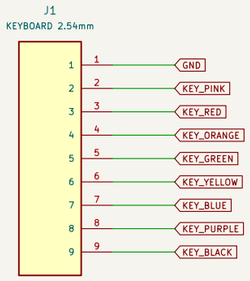
DJTable — Raspberry Pi Zero 2 W

Sheet: /
File: DJTable.kicad_sch

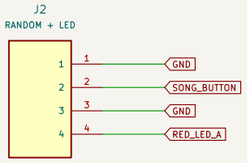
Title: Pi and power distribution

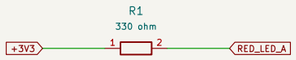
Size: A3	Date: 2026-09-26	**Rev: A0.4**
KiCad E.D.A. 10.0.6		Id: 1/4

EXTERNAL KEYBOARD: EXACT 1x9 ORDER



RANDOM BUTTON + RED LED: EXACT 1x4 ORDER





The LED is in the external button, not another on-board LED.
No LED-control GPIO. Both ground contacts return to GND.

ON-BOARD 6 x 6 mm TACTILE SWITCHES







MODE / SHUTDOWN - FIRMWARE PENDING
Short release: next mode. Hold 3 s: graceful shutdown.
BCM0 is the HAT ID pin. R2 remains DNP initially.
Before fitting R2: disable HAT EEPROM probing, leave
I2C0 unused and enable the shared button handler.
Check boot with button held. Switch power off AFTER halt.

GPIO inputs: enable Pi internal pull-ups in firmware.
No external resistor arrays fitted. All switches close to GND.
Debounce 30 ms. Check tactile switch common legs.

SOURCE-BASED FOOTPRINTS - PHYSICAL FIT CHECK REQUIRED

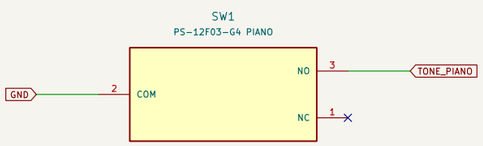
DJTable - Raspberry Pi Zero 2 W

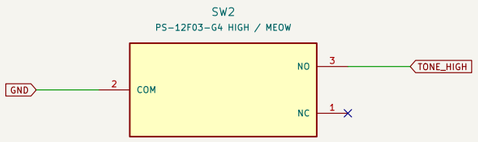
Sheet: /Keyboard, buttons and Random LED/
File: Controls.kicad_sch

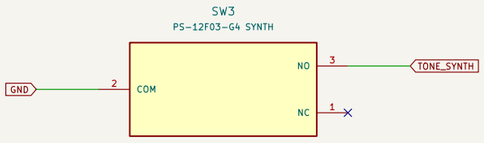
Title: Keyboard, buttons and Random LED

Size: A3	Date: 2026-09-26	**Rev: A0.4**
KiCad E.D.A. 10.0.6		Id: 2/4

MOMENTARY TONE SELECTION

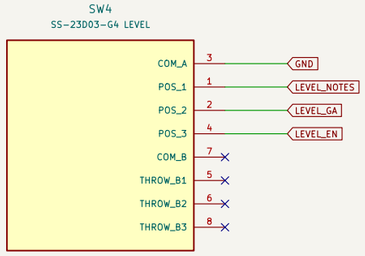


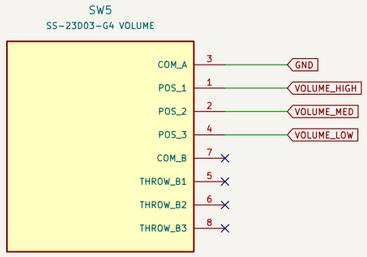




Spring-return: press selects tone, release retains it.
Pins: 1=a NC; 2=b COM; 3=c NO.
Suntan drawing footprint; verify physical fit.

3-POSITION SLIDERS





One stable contact LOW selects each position.
Ignore invalid/multiple contacts while sliding.
Second pole unused. Source drawing: shorting type.
Verify contact truth table on purchased switches.

INTERNAL GPIO PULL-UPS

Use internal GPIO pull-ups; RN1-RN3 removed.
No carrier pull-ups are required for these short switch runs.

SOURCE-BASED FOOTPRINTS - PHYSICAL FIT CHECK REQUIRED

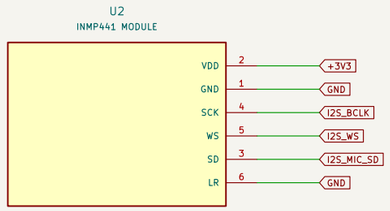
DJTable - Raspberry Pi Zero 2 W

Sheet: /Tone buttons, level and volume/
File: Selectors.kicad_sch

Title: Tone buttons, level and volume

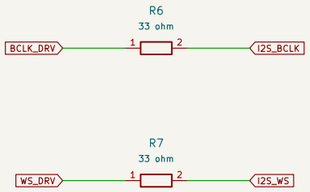
Size: A3	Date: 2026-09-26	**Rev: A0.4**
KiCad E.D.A. 10.0.6		Id: 3/4

INMP441: COMPLETE SOLDERED MODULE

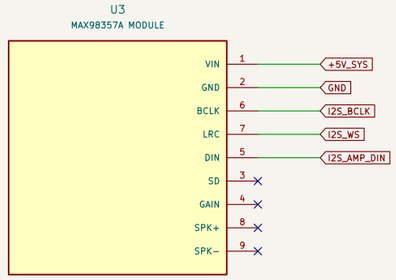


L/R grounded = left slot. 3.3 V only.
BCM20 is dedicated capture data.
Keep sound port clear; clean local ground return.

C3 removed; C4 retained as extra sensor supply bypass.
C5 remains as extra amplifier supply bulk capacitance.



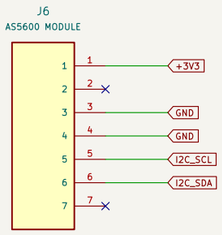
MAX98357A: COMPLETE SOLDERED MODULE



SD and GAIN: no carrier connections; retain module defaults.
Speaker connects to module SPK+ / SPK− only.
Complete module retains its onboard passives.

NO CARRIER MUTE CIRCUIT
U1 pin 7 / BCM4 is unused. U3 SD and GAIN are unconnected.
Complete Adafruit-compatible module supplies SD bias.
No software-controlled amplifier shutdown.

PLATTER: EXACT 1x7 ORDER



VCC=3.3 V; I2C address 0x36.
OUT and GPO/PGO are unused.
No OTP/programming connection.
Use existing Pi I2C pull-ups; check parallel pull-ups on the module.





SHARED I2S
48 kHz; two 32-bit slots; 64 BCLK/frame.
BCLK 3.072 MHz. Pi drives both clocks.
No INMP/amp data pins tied together.

SOURCE-BASED FOOTPRINTS - PHYSICAL FIT CHECK REQUIRED
DJTable - Raspberry Pi Zero 2 W

Sheet: /I2S audio modules and AS5600 connector/
File: Audio_Sensor.kicad_sch

Title: I2S audio modules and AS5600 connector

Size: A3	Date: 2026-09-26	**Rev: A0.4**
KiCad E.D.A. 10.0.6		Id: 4/4